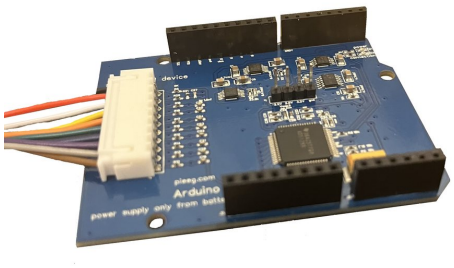
PRODUCT DATASHEET · SKU CQA240229AR

8-Channel EEG Shield for Arduino UNO R4 WiFi

An open-source Arduino shield that turns an Arduino UNO R4 WiFi into an 8-channel biosignal recorder. A Texas Instruments ADS1299 reads EEG, EMG and ECG at 24 bits, and the Arduino streams the data over Wi-Fi to a Python script on your computer. The lowest-cost way into the PiEEG family.



8	24-bit	250 SPS	0.35 µV	Wi-Fi
Channels + REF + BIAS	TI ADS1299 front end	All channels together	RMS noise, 1–40 Hz	UDP to your computer

01 OVERVIEW

Key features

- **Plugs onto the Arduino**: standard UNO header layout and mounting holes, stacking headers.
- **Research-grade front end**: TI ADS1299, 8 simultaneous 24-bit channels, programmable gain 1–24.
- **Wireless data**: the UNO R4 WiFi sends 50-sample UDP packets to any computer on the same network.
- **Low noise**: about 0.35 µV RMS per channel in the 1–40 Hz band (inputs shorted, measured).
- **Free pins**: only D7 and D10–D13 are used; the rest drive LEDs, motors, buzzers or sensors.
- **Open software**: MIT-licensed Arduino sketch, Python receiver, datasets and a Colab manual.

Typical applications

- STEM and neuroscience teaching
- Arduino BCI projects: LEDs, cars, robot arms
- Alpha, blink and chewing experiments
- EMG and ECG experiments
- Embedded signal processing
- Low-cost lab demonstrators

02 TECHNICAL SPECIFICATIONS

ANALOG ACQUISITION	
Channels	8 biosignal inputs
Extra inputs	Reference (SRB1, shared by all channels) and bias drive
Analog front end	Texas Instruments ADS1299
Resolution	24-bit delta-sigma, all channels sampled at the same moment
Sample rate	250 SPS set by the sketch (ADS1299 supports 250 SPS to 16 kSPS)
Gain	1 set by the sketch (ADS1299 supports 1, 2, 4, 6, 8, 12, 24)
Reference voltage	4.5 V internal; 1 LSB ≈ 0.268 µV at gain 1
Internal noise	≈ 0.5 µV RMS raw, ≈ 0.35 µV RMS in 1–40 Hz (inputs shorted, all 8 channels)
Lead-off	Supported by the ADS1299 (impedance check); off in the standard sketch
Signals	EEG, EMG, ECG
HOST INTERFACE	
Host	Arduino UNO R4 WiFi (the Minima has no Wi-Fi)

For research and education only. Not a medical device. Specifications subject to change.

Connectors	UNO-layout stacking headers: 10 + 8 digital, 8 power, 6 analog
Bus	SPI on D13 / D12 / D11, chip select D10, mode 1 · DRDY on D7 (active low)
Wireless link	Wi-Fi 2.4 GHz, UDP port 13900, 1,350-byte packets (50 samples), 5 per second
Indicators	LED D1 power · LED D2 ADS1299 connection
ELECTRODES	
Electrode type	Dry or wet Ag/AgCl
Connector	1 × 10 shrouded header, 2.54 mm: E1–E8, Biasout, Ref
POWER & SAFETY	
Supply	5 V through the Arduino, from a small power bank: ≤ 1 A, ≤ 1,000 mAh
Isolation	None. Battery operation of the whole setup is required
Classification	Research and education tool. Not a medical device
MECHANICAL	
PCB	68.6 × 53.3 mm, 2-layer FR-4, Arduino UNO outline, blue mask
Mounting	4 × Ø3.2 mm holes on the Arduino UNO pattern
Shipping weight	100 g (packed)
SOFTWARE	
Arduino	`1.Send_data_via_wifi.ino` — ADS1299 set-up, sampling, Wi-Fi UDP
Python	`1.Alpha_real_time.py` live 8-channel graphs · `speed_test_250_samples.py`

03 BLOCK DIAGRAM

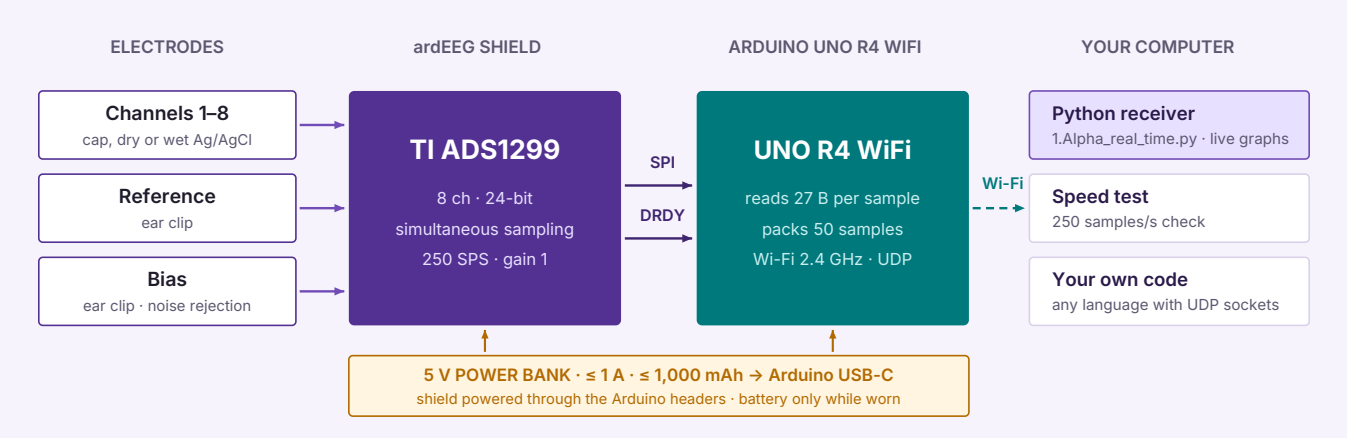


Figure 1 — Signal path. The ADS1299 digitises all channels together; the Arduino reads each sample when DRDY goes low and sends 50 samples per Wi-Fi packet.

04 MEASURABLE SIGNALS

Signal	What it shows	Typical first test
EEG	Brain rhythms: delta, theta, alpha, beta, gamma	Close the eyes and watch alpha (8–12 Hz) rise
EMG	Muscle activity	Chew or clench the jaw
ECG	Heart activity	Electrodes on the chest or wrists

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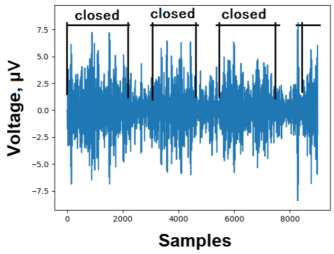


Figure 2 — Alpha test, dry electrode at Fz, 8–12 Hz.

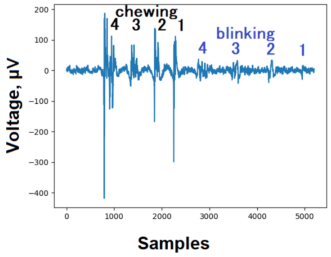


Figure 3 — Chewing and blinking, 1–40 Hz.

05 PIEEG FAMILY

	ardEEG	PiEEG-8	IronBCI-8
Channels	**8**	8	8
Host	**Arduino UNO R4 WiFi**	Raspberry Pi 4 / 5	Built in (BLE)
Link to computer	**Wi-Fi (UDP)**	Software on the Pi	Bluetooth LE
Power	**5 V, ≤ 1 A bank**	5 V, 3 A bank	LiPo
List price	**USD 350**	USD 350	USD 420

06 ORDERING INFORMATION

Product	ardEEG — 8-channel EEG shield for Arduino UNO R4 WiFi
SKU	CQA240229AR
Price	USD 350 (list)
Purchase	Elecrow (official manufacturing partner) — elecrow.com/ardeeg.html · pieeg.com/ardeeg

Included	
1 × ardEEG shield	ADS1299, stacking headers, electrode header

Not included — required	
Arduino UNO R4 WiFi	WiFi version
5 V power bank	≤ 1 A, ≤ 1,000 mAh
USB-C cable	Sketch upload, then power
Electrodes and cap	8 Ag/AgCl + 2 ear clips
Computer	Python 3, same 2.4 GHz Wi-Fi

WARNING Battery only.

ardEEG has no isolation. While electrodes are on a person the Arduino runs from a small power bank only — never from a computer's USB port or a wall charger. ardEEG is not a medical device.

07 RELATED DOCUMENTS

AE-MD-002 Mechanical & Dimensional Drawing	**AE-PCB-003** PCB & Manufacturing Specification	**AE-PIN-004** Connector Pinout & Electrode Guide
AE-QSG-005 Quick Start & Safety Guide	**AE-SW-006** Data Interface & Software Guide	

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